

FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10: CHEMICALS OF LIFE (FINAL EXPLANATION)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have finished all 6 lessons on the Chemicals of Life. Now write your own COMPLETE EXPLANATION of this driving question: How do the simple foods Kenyan athletes eat, like ugali, githeri and beans, provide the chemicals their bodies need for energy, growth and repair? Think about Eliud Kipchoge, who trains near Iten in the Rift Valley and ran 42.195 km in 7,269 s (2:01:09) at Berlin 2022. Use your class model (Lesson 1 and final), your Driving Question Board, your food test results, the amylase data and the meal comparison. Write your answer in the space under each prompt. Use numbers from the tables whenever you make a claim. Do not call a meal 'best' without citing numbers. Remember safety: goggles and a hot water bath for Benedict's and HCl tests, and never taste lab samples. Use the words from the unit, such as reducing sugar, amino acids, fatty acids and glycerol, optimum and denatured. Write in full sentences in your own words.

Part 1: The plain plate and the marathon

Look at Table 1 and the Kipchoge figures. Kipchoge has a body mass of about 52 kg and uses about 1 kcal per kg per km. His marathon (42.195 km) takes 7,269 s. (a) Calculate his average pace in s per km and the energy cost of the marathon. (b) His stored glycogen is about 500 g. Using 4 kcal per g, calculate the energy it holds and compare it with the marathon cost. (c) Name the three main groups of chemicals of life taught in this unit, and give the one-line role of each in the body. (d) Explain why a plate of ugali, beans, sukuma wiki and milk is 'not plain'.

Write your answer here:

Part 2: Food tests as evidence

Use your test results from Lessons 2, 3 and 4. (a) Complete the table below by writing the reagent or test used and the colour or sign that shows a positive result for each chemical.

Chemical	Test or reagent	Positive result
Starch		
Reducing sugar		
Protein		
Lipid		

(b) Sugarcane juice tastes very sweet but Benedict's solution stayed blue. Explain this result and describe what you did to get a positive result. (c) Ugali stayed blue with Biuret reagent, but beans turned violet. Explain what this tells you and what it does NOT tell you about ugali. (d) Give one food that tested positive for lipids and one test result that shows maize flour has no lipid.

Write your answer here:

Part 3: Enzymes and digestion

Use the amylase data in Table 3 (1% starch solution plus saliva amylase; the time for iodine to stop turning blue-black).

Temperature	Time to clear starch	Relative rate (60 / time)
0 C	not cleared in 600 s	0
20 C	300 s	0.2
37 C	60 s	1.0
60 C	not cleared in 600 s	0
80 C	not cleared in 600 s	0

(a) Describe the pattern in the data and name the optimum temperature. (b) Calculate how many times faster the reaction is at 37 C than at 20 C. (c) Explain why no starch was cleared at 60 C and 80 C, using the word 'denatured'. (d) At pH 6.8 the rate was 1.0 but at pH 2 and pH 12 it was 0. What does this show about pH? (e) What is an enzyme, what does amylase break starch into, and why does chewed ugali taste sweet?

Write your answer here:

Part 4: Which meal best supports the athlete?

Mama Wanjiru, a canteen cook in Eldoret, must choose a meal for young runners from Iten. Use the Atwater factors ($\text{Energy} = 4 \times \text{C} + 4 \times \text{P} + 9 \times \text{F}$) and the Table 4 data below.

Meal	Foods	Carbohydrate (g)	Protein (g)	Fat (g)	Energy (kcal)
A	Ugali 300 g + beans 150 g + sukuma wiki 100 g + milk 250 g	126.25	30.25	15.3	763.7
B	Githeri 400 g + sukuma wiki 100 g	77	26.5	7	477
C	Mandazi 200 g + nyama choma 100 g	90	39	45	921

(a) Show the energy calculation for Meal C and the percentage of its energy that comes from fat. (b) Which meal has the most carbohydrate, and which has the highest percentage of energy from fat? (c) Write a Claim-Evidence-Reasoning answer to the question: which meal best supports an endurance athlete? Use at least three numbers from the table, and compare the meal with the marathon energy cost of about 2,200 kcal. (d) Explain why a runner should not choose a meal only because it has the most energy.

Write your answer here:

Part 5: Putting it all together

Write a complete final explanation that answers the driving question: How do the simple foods Kenyan athletes eat, like ugali, githeri and beans, provide the chemicals their bodies need for energy, growth and repair? In your answer: (a) name the chemical of life in each food and its role (energy, growth and repair, stored energy and insulation); (b) refer to at least two test results as evidence; (c) explain the part played by enzymes such as amylase and what affects them; (d) use at least one number from Table 1, Table 3 or Table 4; (e) say how your final class model is different from your initial model in Lesson 1.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of chemicals of life	Correctly names carbohydrates, proteins and lipids, gives their building blocks (sugars, amino acids, fatty acids and glycerol) and roles, and links each to Kenyan foods such as ugali, beans, milk and groundnuts.	Names the three groups and gives correct roles for most, with some Kenyan food examples but little detail on structure.	Names some groups but roles or foods are missing, confused or incorrect.
Use of evidence (tests and tables)	Accurately uses food test colours and results (iodine, Benedict's, Biuret, brown paper and emulsion) and specific numbers from Tables 1, 3 and 4 to support every claim.	Uses some test results and numbers correctly, but a few claims have no evidence.	Gives few or incorrect results or numbers; claims are mostly unsupported.
Enzyme explanation	Explains that enzymes are protein catalysts, that amylase breaks starch to maltose, and explains the optimum temperature and pH, denaturation and specificity using the rate data (for example, 5 times faster at 37 C than at 20 C).	Explains the role of amylase and the optimum conditions, with limited use of data or of the term denatured.	States that enzymes help digestion but gives little or incorrect detail on conditions or data.
Explaining the phenomenon	Clearly explains how ugali, beans,	Explains most of how the plate supports	Gives a general statement that food

	sukuma wiki and milk supply energy, repair and stored energy for a marathon runner such as Kipchoge, linking to the 2,200 kcal cost and 2,000 kcal glycogen store.	performance, but the link to the marathon figures is partial.	gives energy without linking chemicals to the phenomenon.
Scientific reasoning (Claim-Evidence-Reasoning)	Makes a clear claim about the best meal, supports it with at least three correct Table 4 numbers, and reasons logically why carbohydrate, protein and fat shares matter.	Makes a claim with some correct numbers and partial reasoning.	Makes a claim with no numbers or reasoning, or the numbers are wrong.

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